Exercise 1

a) The series is divergent and therefore also absolutely divergent since the n-th term doesn't go to 0.
In fact $\lim_{n\to\infty} \left|\frac{2^n}{\sqrt{n}}\right| = \lim_{n\to\infty} \frac{|2|^n}{\sqrt{n}} = +\infty$

by a proposition seen in class (exponential beats power)

b) The series becomes $\sum_{n=1}^{\infty} \frac{4}{\sqrt{n}} = \sum_{n=1}^{\infty} n^{-1/2}$
This is divergent ~~and~~ absolutely divergent because it's a series of the form $\sum_{n=1}^{\infty} n^{\alpha}$ and $\alpha \geq -1$.

c) Here the series is absolutely convergent and therefore converged.
In fact $\sum_{n=1}^{\infty} \left|\frac{2^n}{\sqrt{n}}\right| \leq \sum_{n=1}^{\infty} \frac{|2|^n}{\sqrt{n}} \leq \sum_{n=1}^{\infty} |2|^n$ which is the geometric series and converges since $|2| < 1$

d) Here the series in convergent but not absolutely convergent.
In fact $\sum_{n=1}^{\infty} \left|\frac{2^n}{\sqrt{n}}\right| = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ which diverges as seen in b)
Instead $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ is convergent by Leibnitz criterion since $\left(\frac{1}{\sqrt{n}}\right)$ is decreasing and $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$.

e) It's divergent and absolutely divergent as seen in a).

Exercise 2

a) Notice that $c_k = \sum_{\substack{l+m=k \\ l,m\geq 0}} a_l b_m$

Then, $C_n = \sum_{k=0}^{n} c_k = \sum_{\substack{k\leq n \\ k\geq 0}} \sum_{\substack{l+m=k \\ l,m\geq 0}} a_l b_m$

$$= \sum_{\substack{l+m\leq n \\ l,m\geq 0}} a_l b_m$$

At the same time $A_n B_n = \left(\sum_{0\leq l\leq n} a_l\right)\left(\sum_{0\leq m\leq n} b_m\right)$

$$= \sum_{\substack{l,m\leq n \\ l,m\geq 0}} a_l b_m$$

There, noticing that $l+m\leq n \Rightarrow l,m\leq n$, the second sum is larger than the first one, sicence they are both sums of positive terms and in the second one the indeces are ranging on a bigger set. Hence $C_n \leq A_n B_n$

b) From a) we can deduede, replacing n with $2n$, that

$C_{2n} = \sum_{\substack{l+m\leq 2n \\ l,m\geq 0}} a_l b_m$

Now we notice that $l,m\leq n \Rightarrow l+m\leq 2n$ and then, note it's can the sum where the indeces range on a bigger set than in $A_n B_n$. Hence $C_{2n} \geq A_n B_n$.

c) Since $a_n, b_n \geq 0$, $A_n \leq \sum_{k=0}^{\infty} a_k$ and $B_n \leq \sum_{k=0}^{\infty} b_k$.
Also $c_n \geq 0$, hence C_n is an increasing sequence.
From a) we know that $C_n \leq A_n B_n \leq \left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) < +\infty$.
Hence C_n is increasing and bounded, therefore convergent.
In other words $\sum_{n=0}^{\infty} c_n = \lim_{n\to\infty} C_n < +\infty$.
Finally combining a) and b)

$C_n \leq A_n B_n \leq C_{2n}$ and taking the limit,

$$\sum_{n=0}^{\infty} c_n \leq \left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) \leq \sum_{n=0}^{\infty} c_n$$

Then $\sum_{n=0}^{\infty} = \left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right)$.

Exercise 3

a) Let us call $A_n = \sum_{e=0}^{n} a_e$ and $A'_n = \sum_{e=0}^{n} a_{\sigma(e)}$

$$A'_n = \sum_{e\in\{0,\dots,n\}} a_{\sigma(e)} = \sum_{s\in\{\sigma(0),\dots,\sigma(n)\}} a_s = \sum_{s\in\sigma(\{0,\dots,n\})} a_s$$

here we use that $\sigma(n)$ is injective.

Now we notice that $\{0,\dots,n-k\} \subseteq \sigma(\{0,\dots,n\}) \subseteq \{0,\dots,n+k\}$.
the second containment is easier to prove and follows from the fact that $\sigma(e) \le e+k$
For the first one suppose (by contradiction) that $\exists\, e \le n-k$ s.t. $e \notin \sigma(\{0,\dots,n\})$.
By surjectivity of σ $\exists\, m > n$ s.t. $\sigma(m) = e$ but
$|\sigma(m) - m| = m - \sigma(m) > n-(n-k) = k$ contradiction.

Now

$$\cancel{A_{n-k} = \sum_{e\in\{0,\dots,n-}}}$$

$$|A'_n - A_n| = \left|\sum_{s\in\sigma(\{0,\dots,n\})} a_s - \sum_{s\in\{0,\dots,n\}} a_s\right|$$

$$\le \left|\sum_{s\in\sigma(\{0,\dots,n\})\setminus\{0,\dots,n\}} a_s - \sum_{s\in\{0,\dots,n\}\setminus\sigma(\{0,\dots,n\})} a_s\right|$$

$$\le \sum_{s\in\{n+k,\dots,n+1\}} |a_s| + \sum_{s\in\{n-k+1,n\}} |a_s|$$

$$\le |a_{n-k+1}| + \dots + |a_{n+k}|$$

$$\lim_{n\to\infty} |A'_n - A_n| = \lim_{n\to\infty} |a_{n-k+1}| + \dots + \lim_{n\to\infty} |a_{n+k}| = 0$$

Hence $\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} a_{\sigma(n)}$

b) Take the following rearrangement

$$\underbrace{-1, \tfrac{1}{2}, -\tfrac{1}{3}}_{0\text{-th}}, \underbrace{\tfrac{1}{4}}_{1\text{st}}, -\tfrac{1}{5}, \underbrace{\tfrac{1}{6}, \tfrac{1}{8}}_{2\text{-nd}}, -\tfrac{1}{7}, \underbrace{\tfrac{1}{10}, \dots, \tfrac{1}{16}}_{3\text{-rd}}, -\tfrac{1}{9},$$
$$\underbrace{\tfrac{1}{18}, \dots, \tfrac{1}{32}}_{4\text{-th}}, -\tfrac{1}{11}, \underbrace{\tfrac{1}{34}, \dots, \tfrac{1}{64}}_{5\text{-th}}, -\tfrac{1}{13}$$

we put $-1, \frac{1}{2}, -\frac{1}{3}$ in the 0-th block.
Then in the n-th block, for $n > 1$, we put the even terms from $\frac{1}{2^n}$ excluded to $\frac{1}{2^{n+1}}$ included followed by $-\frac{1}{2n+3}$.

It's easy to see that the rearrangement is actully a bijection since every term is taken exactly in one block (check for odd and even separately)

Now we have to check that it is divergent

Let's call n_k the index at the end of the k-th block

And set $S_n = \sum_{k=1}^{n} a_{\sigma(k)}$ where $a_{\sigma(k)}$ is our rearrangement

For instance $n_0 = 3$ and $S_{n_0} = S_3 = -1 + \frac{1}{2} = \frac{1}{3}$.

We notice that $S_n \geq S_{n_k}$ $\forall n \geq n_k$ and $k \geq 1$

This is because every term of the block is positive until $n_{k+1} - 2$ hence S_n increases and finally $-\frac{1}{2n+3}$ is bigger then $-\frac{1}{4}$, while $S_{n_{k+1}-2} - S_{n_k} \geq \frac{1}{4}$ because it's given by the sum of 2^{n-1} terms, bigger than $\frac{1}{2^{n+1}}$.

By induction $S_n \geq S_{n_k}$ $\forall$ $n \geq n_k$

Now we just need to show that S_{n_k} diverges to $+\infty$

We just prove it is monotone then to prove that it is unbounded we just need to prove that is not convergent.

In fact $|S_{n_{k+1}} - S_{n_k}|$ doesn't go to 0 and this will conclude the proof.

Indeed $S_{n_{k+1}} - S_{n_k} = \underbrace{\frac{1}{2^k+2} + \dots + \frac{1}{2^{k+1}}}_{\geq \frac{1}{4}} - \underbrace{\frac{1}{2k+3}}_{\leq \frac{1}{5}} \geq \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$.

Exercise 4

a) $\lim_{x \to x_0} = x_0$. Fix $\varepsilon > 0$

~~Look at $|x - x_0| < \varepsilon$~~ and pick $\delta = \varepsilon$

We get that $|x - x_0| < \varepsilon$ whenever $|x - x_0| < \delta = \varepsilon$ and $x \neq x_0$.

as whished.

b) $\lim_{x \to 3} x^2 = 9$

Fix $\varepsilon > 0$

We try and solve $|x^2 - 9| < \varepsilon$.

For $x \geq 3$: $x^2 - 9 < \varepsilon \iff x^2 < 9 + \varepsilon \iff x < \sqrt{9+\varepsilon}$

For $0 \leq x \leq 3$: $-x^2 + 9 < \varepsilon \iff x^2 > 9 - \varepsilon \iff x > \sqrt{9-\varepsilon}$

So for $x \in (\sqrt{9-\varepsilon}, \sqrt{9+\varepsilon})$ we have $|x^2 - 9| < \varepsilon$

Notice that $\sqrt{9-\varepsilon} < 3 < \sqrt{9+\varepsilon}$ and set $\delta = \min\{\sqrt{9+\varepsilon} - 3, 3 - \sqrt{9-\varepsilon}\}$

Then $|x^2 - 9| < \varepsilon$ whenever $|x - 3| < \delta$ and $x \neq 0$ $\square$

c) By operations of limits

$$\lim_{x \to 1} \frac{x^2+3}{x+1} = \frac{\left(\lim_{x \to 1} x\right)^2 + 3}{\lim_{x \to 1} x + 1} \quad \text{using b)}$$

$$= \frac{1^2 + 3}{1 + 1} = 2$$

d) We first compute

$\lim_{x \to 0} x^2 + x = \left(\lim_{x \to 0} x\right)^2 + \lim_{x \to 0} x = 0^2 + 0 = 0$, $\lim_{x \to 0} -(x^2 + x) = 0$

And since $-1 < \sin(1/x) \leq 1$ we have $-(x^2+x) \leq (x^2+x)\sin(1/x) \leq x^2 + x$

By squeeze thm $\lim_{x \to 0} (x^2+x)\sin(1/x) = 0$.

e) Define $a_n = \begin{cases} 1/n & n \text{ even} \\ \sqrt{2}/n & n \text{ odd} \end{cases}$

We easily check that $|a_n| \leq \frac{\sqrt{2}}{n}$ $\lim_{n \to \infty} a_n = 0$ and $a_n \neq 0$

$f(a_n) = \begin{cases} 1 & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$

Hence $\lim_{n \to \infty} f(a_n)$ does not exist $\Rightarrow \lim_{x \to 0} f(x)$ does not exist

by the characterization of limit ~~seq~~ sequences seen in class

Exercise 5

a) Fix $\varepsilon > 0$

By def of limit:

$\exists\ \delta_1 > 0$ s.t. $|f(x) - a| < \varepsilon \quad \forall x : |x - x_0| < \delta_1 \quad x \neq x_0$

$\exists\ \delta_2 > 0$ s.t. $|g(x) - a| < \varepsilon \quad \forall x : |x - x_0| < \delta_2 \quad x \neq x_0.$

Take $\delta = \min\{\delta_1, \delta_2\}$

Notice that $|\max\{f(x), g(x)\} - a| \leq |f(x) - a|$ or $|\max\{f(x), g(x)\} - a| = |g(x) - a|$

Hence $|\max\{f(x), g(x)\} - a| < \varepsilon$ whenever $|x - x_0| < \delta$ and $x \neq x_0$

b) Take $\varepsilon = \frac{a-b}{3}$

By def of limit

$f(x) > a - \varepsilon$

$\exists\ \delta_1 > 0$ s.t. $|f(x) - a| < \varepsilon \quad \forall x : |x - x_0| < \delta_1 \quad x \neq 0$

$\exists\ \delta_2 > 0$ s.t. $|g(x) - b| < \varepsilon \quad \forall x : |x - x_0| < \delta_2 \quad x \neq 0$

$g(x) < b + \varepsilon$

Take $\delta_0 = \min\{\delta_1, \delta_2\}$

Then $\forall x$ s.t. $|x - x_0| < \delta_0 \quad f(x) > a - \varepsilon > b + \varepsilon > g(x)$

c) From b) when $a > b$

$\lim_{x \to x_0} \max\{f(x), g(x)\} = \lim_{x \to 0} f(x) = a = \max\{a, b\}$

Since f and $\max\{f, g\}$ coincide on $(x_0 - \delta, x_0 + \delta) \setminus \{x_0\}$ when $\delta < \delta_0$.

One can obtain the same conclusion when $b > a$.

Combing this with a) we get that, in every case,

$\lim_{x \to x_0} \max\{f(x), g(x)\} = \max\{a, b\}.$

To prove this. Fix $\varepsilon > 0$

get δ' from def of limit of f at x_0

Use $\delta = \min\{\delta', \delta_0\}$ to get the def of $\max\{f, g\}$ having limit a at x_0.